# APPENDIX F

# Laplace transforms

## F.1. Using Laplace transforms

Laplace transforms are a convenient way to solve time-invariant linear equations. These often arise in control-system design. In many cases the full nonlinear system can be linearized about an operating point and the resulting system is often time invariant. Laplace transforms can be used for such systems.

The definition of the one-sided Laplace transform is

$$F(s) = \{f(t)\} = \int_{0-}^{\infty} f(t)e^{-st}dt \tag{F.1}$$

The lower integral limit, 0– means that it includes 0 so that impulses may be modeled. An impulse is an input of infinitesimal duration. For solving differential equations the most important properties are differentiation.

$$\dot{f} = sF(s) - f(0-) \tag{F.2}$$

and second differentiation

$$\ddot{f} = s^2F(s) - sf(0-) - \dot{f}(0-) \tag{F.3}$$

For example, assume that we have the second-order differential equation

$$\ddot{x} + x = u \tag{F.4}$$

$$x(0) = x_0 \tag{F.5}$$

$$\dot{x}(0) = \dot{x}_0 \tag{F.6}$$

The Laplace transform of the differential equation is

$$s^2X(s) - sx(0-) - x(\dot{0}-) + X(s) = u(s) \tag{F.7}$$

or

$$X(s) = \frac{u(s)}{s^2+1} + \frac{sx_0 + \dot{x}_0}{s^2+1} \tag{F.8}$$

There are two parts to the response. One is due to the initial conditions, $x(0)$ and $\dot{x}(0)$, and the other due to the input.

Two special inputs are the step and the impulse. In the case of a step

$$u(s) = \frac{u_0}{s} \tag{F.9}$$

With a step, the input is at a constant value from time 0 to time infinity. For an impulse

$$u(s) = u_0 \tag{F.10}$$

The input is nonzero only at time 0. An impulse (sometimes called the Dirac-delta function, $\delta$), is an input with an amplitude of $u_0$ but that happens in an infinitesimally short time. In this case the step response with zero initial conditions is

$$X(s) = \frac{u_0}{s(s^2+1)} \tag{F.11}$$

and the impulse response is

$$X(s) = \frac{u_0}{s^2+1} \tag{F.12}$$

Using a table of Laplace transforms the solution for the impulse response in the time domain is

$$x(t) = u_0 \sin t \tag{F.13}$$

For the step we first expand into partial fractions

$$X(s) = \frac{as+b}{s^2+1} + \frac{c}{s} \tag{F.14}$$

The numerator is always the order $s$ one less than the denominator.

We need to match terms so we put this over a common fraction

$$X(s) = \frac{cs^2 + c + as^2 + bs}{s(s^2+1)} \tag{F.15}$$

Matching terms

$$c = u_0 \tag{F.16}$$

$$a = -u_0 \tag{F.17}$$

$$b = 0 \tag{F.18}$$

or

$$X(s) = \frac{u_0}{s} - \frac{su_0}{s^2+1} \tag{F.19}$$

Again using a table of transforms, the solution for the step input is

$$x(t) = u_0(1 - \cos t) \tag{F.20}$$

The first term comes from the $\frac{u_0}{s}$ and the second from the $\frac{su_0}{s^2+1}$ term. We could have solved for the response to the initial conditions the same way. Usually, we use the Laplace transform to solve for the response to inputs, but as shown above it can always be used to solve the response to both initial conditions and inputs.

## F.2. Useful transforms

Table F.1 gives a short list of Laplace transform pairs useful for control work.

**Table F.1** Laplace transform pairs.

| **Transform Pair** | $F(s)$ | $f(t)$ |
|---|---|---|
| Single Differentiation | $sF(s) - f(0+\epsilon)$ | $\frac{df(t)}{dt}$ |
| Multiple Differentiation | $s^n F(s) - \sum_{k=1}^{n} s^{n-k} \frac{d^{k-1}f}{dt^{k-1}}(0+\epsilon)$ | $\frac{d^n f(t)}{dt^n}$ |
| Integration | $\frac{F(s)}{s} + \frac{\int f(t)dt}{s}$ | $\int f(t)dt$ |
| Linearity | $aF(s)$ | $af(t)$ |
| Time translation | $e^{\tau s}F(s)$ | $f(t+\tau)$ |
| S translation | $F(s+\frac{1}{T})$ | $e^{\frac{-t}{T}}f(t)$ |
| Impulse | 1 | $\delta(t)$ |
| Doublet | $s$ | $\dot{\delta}(t)$ |
| Step | $\frac{1}{s}$ | 1 |
| Lag | $\frac{T}{Ts+1}$ | $e^{-\frac{t}{T}}$ |
| Sine | $\frac{\omega_n}{s^2+\omega_n^2}$ | $\sin \omega_n t$ |
| Cosine | $\frac{s}{s^2+\omega_n^2}$ | $\cos \omega_n t$ |
| Second-order sine | $\frac{1}{s^2+2\zeta\omega+\omega_n^2}$ | $\frac{1}{\omega_n\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2}\, t$ |
| Second-order cosine | $\frac{s}{s^2+2\zeta\omega+\omega_n^2}$ | $\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2}\, t+\theta)$ |